

Scalability & Future Plan

Date	01 July 2026
Team ID	SWTID-2026-4350
Project Name	Credit Card Approval Prediction
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	The system predicts loan approval using only the available applicant information and trained ML model.	Prediction accuracy depends on the quality of the training data.	High
2	The application is currently designed for local deployment only.	Users cannot access the system remotely over the internet.	Medium
3	The system does not integrate with real-time banking or credit bureau APIs.	Financial information cannot be verified automatically.	Low

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Suitable for a limited number of users.	Deploy the application on cloud platforms (AWS, Azure, Render) with load balancing to support more users.
2	Data Storage	Uses a local SQL database.	Migrate to cloud databases such as MySQL or PostgreSQL for better scalability and reliability.
3	Performance	Predictions are generated using a single ML model.	Optimize the model, improve preprocessing, and implement caching for faster response times.
4	Security	Basic authentication and local database storage.	Implement secure authentication, password encryption, HTTPS, and role-based access control.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 2	Improve the Machine Learning model by retraining with more applicant data to increase prediction accuracy.	1–2 Months	More accurate loan approval predictions.
Phase 3	Enhance the dashboard with additional analytics such as approval trends, rejection statistics, and applicant reports.	1–2 Months	Better monitoring and decision-making for administrators.
Phase 4	Optimize database performance and improve application response time for handling larger datasets.	1–2 Months	Better scalability and faster performance.